

Notes: Menzio & Shi (JPE, 2011)

Efficient Search on the Job and the Business Cycle

Contents

1	Why this paper? The empirical motivation	3
1.1	Table 1: what the U.S. data say (stylized facts)	3
2	Environment and timing	3
2.1	Agents and primitives	3
2.2	Within-period timing (conceptual)	4
3	Model 1: the planner's problem (efficient allocation)	5
3.1	State variables	5
3.2	Planner controls	5
3.3	Per-period surplus and laws of motion (Eqs. (1)–(3))	5
3.4	Dynamic programming (Eq. (4))	6
3.5	Theorem 1 (separability) and component value functions (Eqs. (5)–(6))	6
3.6	Policy characterization (Eqs. (7)–(11))	7
3.6.1	Match creation: reservation signal rules (Eqs. (7)–(8))	7
3.6.2	Vacancy creation: optimal tightness (Eqs. (9)–(10))	7
3.6.3	Match destruction: a reservation productivity cutoff (Eq. (11))	7
4	Model 2: decentralization with directed search and contracts	8
4.1	Submarkets and contracts	8
4.2	Worker problem and the “directed search” operator (Eq. (12))	8
4.3	Match value (joint surplus) and on-the-job search (Eq. (14))	8
4.4	Vacancy posting and free entry (Eq. (16))	8
4.5	Block-recursive equilibrium (BRE) and efficiency (Theorem 2)	9
5	Calibration (Table 2 and Fig. 1)	9
5.1	Functional forms and normalization	9
5.2	Table 2: what differs across model variants	9
5.3	Fig. 1: tenure distribution as a key target	10
6	Experience-goods economy ($\alpha = 0$): mechanism and quantitative results	10
6.1	Key qualitative structure	10
6.2	Figures 2–4: impulse responses to a permanent 1% increase in y	11
6.3	Table 3: simulated business-cycle moments (experience model)	11

7	Why on-the-job search and match heterogeneity matter (Table 4)	12
7.1	P-00 benchmark (Pissarides 2000): no OJS, no heterogeneity	12
7.2	MP-94 benchmark (Mortensen–Pissarides 1994): heterogeneity but no OJS	12
8	Inspection-goods economy ($\alpha = 1$): why amplification disappears	13
8.1	Figures 5–7: impulse responses are small	13
8.2	Analytic intuition: why the UE response is limited	13
9	Short summary	13
10	Conclusion	14
10.1	Summary	14
10.2	What the conclusion implies (interpretation)	14
10.3	Economic intuition: why the model amplifies small productivity shocks	15
10.4	Takeaway	16

TABLE 1
SUMMARY STATISTICS, QUARTERLY U.S. DATA

x	u	v	h^{ue}	h^{eu}	h^{ee}	π
$SD(x)/SD(\pi)$	9.56	10.9	5.96	5.48	5.98	1
$Autocorr(x)$	.872	.909	.822	.698	.597	.760
$Corr(\cdot, x)$:						
u	1	-.902	-.916	.778	-.634	-.283
v	. . .	1	.902	-.778	.607	.423
h^{ue}	. . .	. . .	1	-.677	.669	.299
h^{eu}	. . .	. . .	. . .	1	-.301	-.528
h^{ee}	. . .	. . .	. . .	. . .	1	.208
π	. . .	. . .	. . .	. . .	. . .	1

1 Why this paper? The empirical motivation

The paper is built to explain a well-known tension in labor-search macro:

- Aggregate labor productivity is not very volatile over the business cycle.
- Yet unemployment, vacancies, and worker flows are *very* volatile and strongly cyclical.

1.1 Table 1: what the U.S. data say (stylized facts)

Table 1 reports standard deviations (relative to labor productivity), autocorrelations, and cross-correlations for U.S. quarterly data (1951–2006). The message is stark:

1. **Volatility amplification:** unemployment and vacancies have standard deviations about an order of magnitude larger than labor productivity.
2. **Beveridge curve:** unemployment and vacancies are strongly negatively correlated.
3. **Worker flows are strongly cyclical:**
 - the UE rate (job finding) is strongly *procyclical* (negatively correlated with unemployment),
 - the EU rate (separations) is *countercyclical* (positively correlated with unemployment),
 - the EE rate (job-to-job) is *procyclical*.

Modeling goal: generate these comovements *and* large labor-market fluctuations from relatively small productivity shocks *without* assuming inefficiency in search (search will be efficient).

2 Environment and timing

2.1 Agents and primitives

There is a unit measure of risk-neutral workers and free entry of risk-neutral firms (vacancies). Time is discrete.

Productivity. A worker-firm match with idiosyncratic productivity z produces

$$y + z,$$

where y is an aggregate component (common to all matches) and z is match-specific. Unemployed workers produce $b > 0$ (home production).

Aggregate productivity evolves as a Markov process

$$\hat{y} \sim \Phi(\hat{y} \mid y).$$

Vacancies and directed search. A vacancy costs k per period. Search is *directed*: workers choose which submarket to visit, and firms choose where to post vacancies.

Each submarket is characterized by a vacancy-to-applicant ratio (tightness) θ . If a worker visits a submarket with tightness θ :

- the worker meets a firm with probability $p(\theta)$ (increasing, concave),
- a vacancy meets a worker with probability $q(\theta) = p(\theta)/\theta$ (decreasing).

On-the-job search. Unemployed workers get a search opportunity with probability λ_u each period. Employed workers get a search opportunity with probability λ_e (if they keep their job).

Information at matching. When a worker meets a vacancy, a signal s about match quality is observed. The signal has accuracy parameter $\alpha \in [0, 1]$:

- with probability α , the signal reveals match quality ($s = z$),
- with probability $1 - \alpha$, the signal is uninformative (drawn from an unconditional distribution $f(\cdot)$, independent of z).

The paper focuses on two polar cases:

$$\alpha = 0 \quad (\text{experience goods}), \quad \alpha = 1 \quad (\text{inspection goods}).$$

2.2 Within-period timing (conceptual)

A useful way to organize the model is with four stages each period:

1. **Separation stage:** some matches are destroyed (endogenously and exogenously).
2. **Search stage:** workers choose submarkets; firms post vacancies.
3. **Matching stage:** meetings happen; signal s is realized; match creation depends on acceptance rules.
4. **Production stage:** output is produced; next period's aggregate productivity $\hat{y}$ is drawn.

3 Model 1: the planner's problem (efficient allocation)

The planner is introduced for two reasons:

- it gives a transparent representation of *efficient* allocation,
- the decentralized equilibrium will be shown to implement this planner allocation.

3.1 State variables

Let the state at the beginning of a period be

$$\psi \equiv (y, u, g),$$

where u is unemployment and $g(z)$ is the measure of employed workers currently in matches of type z . Thus $\sum_z g(z) = 1 - u$.

3.2 Planner controls

The planner chooses:

- **Destruction rule:** $d(z) \in [\delta, 1]$ is the probability a match of type z is destroyed this period (lower bounded by exogenous destruction δ).
- **Market tightness:** θ_u for unemployed search; $\theta_e(z)$ for employed workers currently in match z .
- **Creation rules:** $c_u(s) \in [0, 1]$ is the probability of forming a match when an unemployed worker meets a vacancy and observes signal s ; $c_e(s, z) \in [0, 1]$ for employed workers in match z meeting a vacancy and observing s .

3.3 Per-period surplus and laws of motion (Eqs. (1)–(3))

Given state ψ and controls, the planner's per-period objective can be written as

$$F(d, \theta_u, \theta_e, c_u, c_e \mid \psi) = -k \left[\lambda_u \theta_u u + \sum_z (1 - d(z)) \lambda_e \theta_e(z) g(z) \right] + b \hat{u} + \sum_z (y + z) \hat{g}(z), \quad (1)$$

where $\hat{u}$ and $\hat{g}(\cdot)$ are the unemployment rate and match distribution at the *production stage* (after matching).

The law of motion for unemployment is

$$\hat{u} = u \left[1 - \lambda_u p(\theta_u) m_u \right] + \sum_z d(z) g(z), \quad (2)$$

where $m_u \equiv \sum_s c_u(s) f(s)$ is the probability that a meeting results in a match for an unemployed worker (averaging over signals).

The law of motion for the match distribution is

$$\begin{aligned}\hat{g}(z') &= u \lambda_u p(\theta_u) \left[\alpha c_u(z') + (1 - \alpha) m_u \right] f(z') \\ &\quad + \sum_z (1 - d(z)) g(z) \left(1 - \lambda_e p(\theta_e(z)) m_e(z) \right) \mathbb{1}_{\{z'=z\}} \\ &\quad + \sum_z (1 - d(z)) g(z) \lambda_e p(\theta_e(z)) \left[\alpha c_e(z', z) + (1 - \alpha) m_e(z) \right] f(z'),\end{aligned}\tag{3}$$

where $m_e(z) \equiv \sum_s c_e(s, z) f(s)$ is the meeting-to-match probability for an employed worker currently in match z .

3.4 Dynamic programming (Eq. (4))

Let $W(\psi)$ be the planner's value function at the production stage. Then

$$W(\psi) = \max_{d, \theta_u, \theta_e, c_u, c_e} \left\{ F(d, \theta_u, \theta_e, c_u, c_e \mid \psi) + \beta \mathbb{E}[W(\hat{\psi}) \mid \psi] \right\},\tag{4}$$

where $\hat{\psi} = (\hat{y}, \hat{u}, \hat{g})$ and $\hat{y} \sim \Phi(\cdot \mid y)$.

3.5 Theorem 1 (separability) and component value functions (Eqs. (5)–(6))

A central theoretical contribution is that the planner's value is *linear* in the distribution (u, g) :

$$W(\psi) = W(y, u, g) = u W_u(y) + \sum_z g(z) W_e(z, y),$$

for suitable component value functions $W_u(y)$ and $W_e(z, y)$.

Interpretation.

- $W_u(y)$ is the social value (at the production stage) of having one more unemployed worker when aggregate productivity is y .
- $W_e(z, y)$ is the social value of having one more worker employed in a match of idiosyncratic quality z when aggregate productivity is y .

The component Bellman equations are (paper Eqs. (5)–(6)):

$$\begin{aligned}W_u(y) &= \max_{\theta_u, c_u} \left\{ -k \lambda_u \theta_u + [1 - \lambda_u p(\theta_u) m_u] (b + \beta \mathbb{E} W_u(\hat{y})) \right. \\ &\quad \left. + \lambda_u p(\theta_u) \sum_{z'} \left[\alpha c_u(z') + (1 - \alpha) m_u \right] f(z') (y + z' + \beta \mathbb{E} W_e(z', \hat{y})) \right\},\end{aligned}\tag{5}$$

$$\begin{aligned}W_e(z, y) &= \max_{d, \theta_e, c_e} \left\{ -k(1 - d) \lambda_e \theta_e + d(b + \beta \mathbb{E} W_u(\hat{y})) \right. \\ &\quad + (1 - d) \left(1 - \lambda_e p(\theta_e) m_e \right) (y + z + \beta \mathbb{E} W_e(z, \hat{y})) \\ &\quad \left. + (1 - d) \lambda_e p(\theta_e) \sum_{z'} \left[\alpha c_e(z', z) + (1 - \alpha) m_e \right] f(z') (y + z' + \beta \mathbb{E} W_e(z', \hat{y})) \right\}.\end{aligned}\tag{6}$$

3.6 Policy characterization (Eqs. (7)–(11))

The optimal policies take *threshold* forms (reservation rules).

3.6.1 Match creation: reservation signal rules (Eqs. (7)–(8))

The planner creates a match after a meeting if the expected value of the new match is at least as large as the value of the worker's outside option (unemployment or current match).

Unemployed worker meeting a firm. Create a match if

$$b + \beta \mathbb{E}W_u(\hat{y}) \leq \alpha(y + s + \beta \mathbb{E}W_e(s, \hat{y})) + (1 - \alpha) \sum_{z'} f(z')(y + z' + \beta \mathbb{E}W_e(z', \hat{y})). \quad (7)$$

This yields a reservation signal cutoff $r_u(y)$.

Employed worker in match z meeting a firm. Switch to the new match if

$$y + z + \beta \mathbb{E}W_e(z, \hat{y}) \leq \alpha(y + s + \beta \mathbb{E}W_e(s, \hat{y})) + (1 - \alpha) \sum_{z'} f(z')(y + z' + \beta \mathbb{E}W_e(z', \hat{y})), \quad (8)$$

which yields a reservation signal cutoff $r_e(z, y)$.

3.6.2 Vacancy creation: optimal tightness (Eqs. (9)–(10))

Tightness equates the marginal vacancy cost with the marginal gain from increasing meeting probabilities (through $p'(\theta)$). For unemployed search markets:

$$k \geq p'(\theta_u) \sum_{s \geq r_u(y)} \left[\alpha(y + s - b + \beta \mathbb{E}(W_e(s, \hat{y}) - W_u(\hat{y}))) + (1 - \alpha) \sum_{z'} f(z')(y + z' - b + \beta \mathbb{E}(W_e(z', \hat{y}) - W_u(\hat{y}))) \right] f(s), \quad (9)$$

with complementary slackness (equality if $\theta_u > 0$). A similar condition holds for employed search markets:

$$k \geq p'(\theta_e(z)) \sum_{s \geq r_e(z, y)} \left[\alpha(s - z + \beta \mathbb{E}(W_e(s, \hat{y}) - W_e(z, \hat{y}))) + (1 - \alpha) \sum_{z'} f(z')(z' - z + \beta \mathbb{E}(W_e(z', \hat{y}) - W_e(z, \hat{y}))) \right] f(s), \quad (10)$$

again with complementary slackness.

3.6.3 Match destruction: a reservation productivity cutoff (Eq. (11))

The destruction decision compares the value of destroying the match (sending the worker to unemployment, without search this period) to the value of keeping it (and possibly allowing on-the-job search). The optimal destruction rule is a cutoff: there exists $r_d(y)$ such that low- z matches are destroyed and high- z matches are kept.

Key intuition. A rise in y raises the value of keeping a match *for all* z , so the planner tends to destroy *fewer* matches (lower $r_d(y)$). This interacts with on-the-job search to generate strong reallocation dynamics.

4 Model 2: decentralization with directed search and contracts

4.1 Submarkets and contracts

A submarket is indexed by a pair (x, r) :

- x is the promised continuation value (utility) to a worker if hired,
- r is the firm's hiring standard: a match is formed iff the signal $s \geq r$.

Given aggregate state ψ , free entry pins down the tightness $\theta(x, r, \psi)$ in each submarket.

Define

$$m(r) \equiv \sum_{s \geq r} f(s),$$

the probability that the signal passes the hiring standard.

4.2 Worker problem and the “directed search” operator (Eq. (12))

Let $V_u(\psi)$ be the value (utility) of being unemployed at the production stage. Define the expected utility gain from visiting submarket (x, r) when current value is V :

$$D(x, r, V, \psi) \equiv p(\theta(x, r, \psi)) m(r) (x - V).$$

Then the unemployed worker's Bellman equation is

$$V_u(\psi) = b + \beta \mathbb{E} \left[\max_{x, r} \left\{ V_u(\hat{\psi}) + \lambda_u D(x, r, V_u(\hat{\psi}), \hat{\psi}) \right\} \middle| \psi \right]. \quad (11)$$

4.3 Match value (joint surplus) and on-the-job search (Eq. (14))

Let $V_e(z, \psi)$ be the *joint* value of a match (worker utility plus firm profit) at production stage. It satisfies

$$V_e(z, \psi) = y + z + \beta \mathbb{E} \left[\max_{d, x, r} \left\{ d V_u(\hat{\psi}) + (1 - d) (V_e(z, \hat{\psi}) + \lambda_e D(x, r, V_e(z, \hat{\psi}), \hat{\psi})) \right\} \middle| \psi \right]. \quad (12)$$

4.4 Vacancy posting and free entry (Eq. (16))

Posting a vacancy in submarket (x, r) yields expected profit

$$\Pi(x, r, \psi) = q(\theta(x, r, \psi)) \sum_{s \geq r} \left(\alpha V_e(s, \psi) + (1 - \alpha) \mathbb{E}_z[V_e(z, \psi)] - x \right) f(s).$$

Free entry implies

$$k \geq \Pi(x, r, \psi), \quad \text{with equality if } \theta(x, r, \psi) > 0. \quad (13)$$

4.5 Block-recursive equilibrium (BRE) and efficiency (Theorem 2)

BRE idea. Even though the full state includes the distribution (u, g) , equilibrium objects can be written as functions of *aggregate productivity* y and match type z only:

$$V_u(\psi) = V_u(y), \quad V_e(z, \psi) = V_e(z, y),$$

and similarly for policy rules. This property is called *block recursivity*.

Why it matters. Block recursivity makes it feasible to analyze business-cycle dynamics with heterogeneity and on-the-job search, because the value functions do not depend on the full high-dimensional distribution.

Efficiency. Theorem 2 in the paper shows that a BRE exists and is *constrained efficient*: the decentralized equilibrium implements the planner allocation. The key reason is that directed search with competitive wage (contract) posting internalizes the matching externality in each submarket.

Proof sketch).

1. Given (r, θ) , free entry pins down the contract payoff x so that firms break even.
2. The unemployed (or employed) worker chooses among submarkets; this selection reproduces the planner's optimal trade-off between (i) high promised value and (ii) high meeting probability.
3. The resulting match value V_e satisfies a contraction mapping (paper Eq. (17)), implying uniqueness; and one can verify that the planner's component values satisfy the same mapping, so they coincide.

5 Calibration (Table 2 and Fig. 1)

5.1 Functional forms and normalization

The paper uses a monthly period and a simple meeting probability:

$$p(\theta) = \min\{\theta^\gamma, 1\}, \quad q(\theta) = \frac{p(\theta)}{\theta}.$$

They normalize $\lambda_u = 1$.

Idiosyncratic productivity z is drawn from a Weibull distribution (then discretized for computation). Aggregate productivity y is estimated from data and approximated by a 3-state Markov chain.

5.2 Table 2: what differs across model variants

Table 2 reports parameter values for four variants:

- **EXP:** experience-goods model ($\alpha = 0$),
- **INS:** inspection-goods model ($\alpha = 1$),

- **P-00:** Pissarides (2000) style benchmark (no on-the-job search and no match heterogeneity),
- **MP-94:** Mortensen–Pissarides (1994) benchmark (heterogeneity but no on-the-job search).

TABLE 2
CALIBRATION OUTCOMES

	Description	EXP	INS	P-00	MP-94
β	Discount factor	.996	.996	.996	.996
b	Home productivity	.907	.716	.710	.739
λ_u	Off-the-job search	1	1	1	1
λ_e	On-the-job search	.735	.904	0	0
γ	Elasticity of ρ with respect to θ	.600	.250	.270	.270
k	Vacancy cost	1.55	2.37	1.85	1.89
δ	Exogenous destruction	.012	.026	.026	.012
μ_z	Average idiosyncratic productivity	0	0	0	0
σ_z	Scale idiosyncratic productivity	.952	.008	0	.467
α_z	Shape idiosyncratic productivity	4	10	. . .	10

NOTE.—Calibrated parameters for different versions of the model. The column EXP refers to the version of the model in which matches are experience goods. The column INS refers to the version of the model in which matches are inspection goods. The column P-00 refers to a version of the experience model in which the parameters λ_u and σ_z are constrained to be equal to zero. The column MP-94 refers to a version of the experience model in which the parameter λ_u is constrained to be equal to zero.

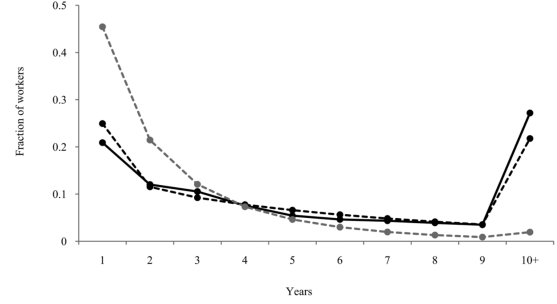


FIG. 1.—Fraction of employed workers who have been on the same job for t years in the data (solid black line), in the experience model (dashed black line), and in the inspection model (dashed grey line).

A helpful way to read Table 2:

1. λ_e governs the intensity of on-the-job search (and is 0 in P-00 and MP-94).
2. (σ_z, α_z) govern heterogeneity in match quality.
3. γ pins down how meeting probabilities respond to tightness (matched to an empirical elasticity of job finding with respect to the vacancy–unemployment ratio).

5.3 Fig. 1: tenure distribution as a key target

Fig. 1 compares the fraction of employed workers who have been in their current job for t years in the data and in the model. The experience-goods version is designed to reproduce the empirical tenure profile via the joint action of:

- endogenous separation of low- z matches,
- on-the-job search out of medium- z matches,
- “absorbing” high- z matches where workers stop searching.

6 Experience-goods economy ($\alpha = 0$): mechanism and quantitative results

6.1 Key qualitative structure

When $\alpha = 0$, workers cannot screen match quality before starting the job (the signal is uninformative). A match’s true quality z becomes relevant for decisions after the worker experiences the job.

In steady state, the planner/equilibrium induces three regions in z :

1. **Low z :** matches are destroyed (worker becomes unemployed).
2. **Middle z :** matches are kept, but the worker searches on the job (job-to-job transitions).
3. **High z :** matches are kept and the worker stops searching (very stable jobs).

This structure is essential for generating job-to-job flows *and* a realistic tenure distribution.

6.2 Figures 2–4: impulse responses to a permanent 1% increase in y

The paper studies a permanent increase in aggregate productivity y and computes the dynamic transition from the initial steady state to the new steady state.

Fig. 2 (flows and unemployment). A positive y shock:

- increases the UE rate (faster job finding),
- decreases the EU rate (fewer separations),
- increases the EE rate (more job-to-job upgrading),
- decreases unemployment substantially.

The economically important point is that the *EE margin is very responsive*.

Fig. 3 (vacancies). Vacancies decompose into postings targeted at unemployed searchers vs employed searchers:

- total vacancies increase,
- vacancies aimed at employed workers increase sharply,
- vacancies aimed at unemployed workers can eventually fall (composition effect as unemployment shrinks).

Thus, the cycle is not only “more vacancies” but also “vacancies reallocated toward poaching employed workers.”

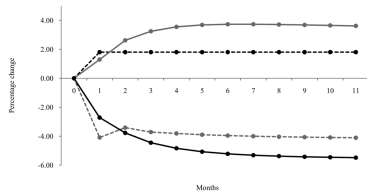


FIG. 2.—Experience model, percentage change of the UE rate (dashed black line), the EU rate (dashed grey line), the EE rate (solid grey line), and the unemployment rate (solid black line) in response to a 1 percent increase in y .

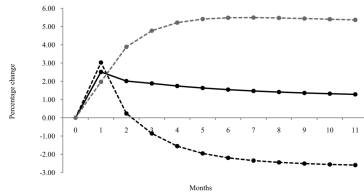


FIG. 3.—Experience model, percentage change of total vacancies (solid black line), vacancies for unemployed workers (dashed black line), and vacancies for employed workers (dashed grey line) in response to a 1 percent increase in y .

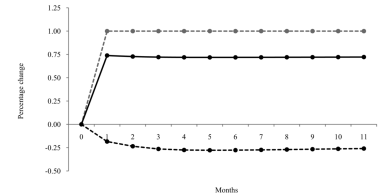


FIG. 4.—Experience model, percentage change of the aggregate component of productivity (dashed grey line), the average idiosyncratic component of productivity (dashed black line), and the average labor productivity (solid black line) in response to a 1 percent increase in y .

Fig. 4 (productivity decomposition). Average labor productivity rises less than the 1% increase in y because average match quality $\bar{z}$ falls:

- the planner destroys fewer low- z matches when y is higher (lower cutoff), which lowers $\bar{z}$,
- this is a composition effect that partially offsets the direct rise in y .

6.3 Table 3: simulated business-cycle moments (experience model)

Table 3 reports moments from a long simulation of the stochastic version of the experience model.

How to read the table.

- The model produces large cyclical movements in unemployment and in the three transition rates (UE, EU, EE).
- Importantly, the signs of the correlations match the data: unemployment is strongly negatively correlated with job finding and job-to-job transitions, and positively correlated with separations.
- Decomposing vacancies into v_u and v_e reveals an additional key prediction:

v_e is strongly procyclical (poaching booms), while v_u can be countercyclical.

This vacancy composition channel is one of the paper’s distinctive mechanisms.

TABLE 3 EXPERIENCE MODEL								
x	u	v	v_u	v_e	h^{uc}	h^{eu}	h^{ee}	π
SD(x)/SD(π)	7.88	2.54	4.29	8.21	2.51	6.23	5.59	1
Autocorr(x)	.850	.637	.748	.824	.799	.772	.823	.762
Corr($\cdot, x$):								
u	1	-.807	.841	-.980	-.976	.972	-.979	-.977
v	. . .	1	-.380	.855	.897	-.898	.858	.894
π	. . .	. . .	-.729	.984	.999	-.979	.983	1

NOTE.—Summary statistics of the last 6,000 months of a 9,000-month-long time series for u , v , v_u , v_e , h^{uc} , h^{eu} , h^{ee} , and π generated by the experience model with aggregate productivity shocks. Section IV provides details on the stochastic process for productivity. All variables are quarterly averages of monthly series. All variables are reported in logs as deviations from an HP trend with smoothing parameter 1,600.

TABLE 4 P-00 AND MP-94 MODELS					
x	u	$v = v_u$	h^{uc}	h^{eu}	π
A. P-00 Model					
SD(x)/SD(π)	.82	2.69	.91	0	1
Autocorr(x)	.815	.677	.994	1	.745
Corr($\cdot, x$):					
u	1	-.932	-.936	0	-.972
v	. . .	1	.990	0	.990
π	. . .	. . .	.999	0	1
B. MP-94 Model					
SD(x)/SD(π)	5.98	4.55	.83	6.61	1
Autocorr(x)	.674	.453	.740	.397	.736
Corr($\cdot, x$):					
u	1	.726	-.737	.906	-.732
v	. . .	1	-.267	.481	-.259
π	. . .	. . .	.998	-.583	1

NOTE.—Summary statistics of the last 6,000 months of a 9,000-month-long time series for u , v , v_u , h^{uc} , h^{eu} , and π generated by a version of the experience model in which the parameters λ_e and σ_e are constrained to be equal to zero. All variables are quarterly averages of monthly series. All variables are reported in logs as deviations from an HP trend with smoothing parameter 1,600.

7 Why on-the-job search and match heterogeneity matter (Table 4)

The paper compares the baseline model to two constrained benchmarks.

7.1 P-00 benchmark (Pissarides 2000): no OJS, no heterogeneity

In P-00, all matches have the same z and $\lambda_e = 0$. Then separations are essentially exogenous and job finding varies little, so unemployment volatility is tiny relative to the data (Table 4, Panel A).

7.2 MP-94 benchmark (Mortensen–Pissarides 1994): heterogeneity but no OJS

In MP-94, matches are heterogeneous in z , but workers cannot search while employed ($\lambda_e = 0$). Table 4, Panel B shows that this benchmark can generate more unemployment variation, but it fails on key comovements (notably the vacancy–unemployment relationship).

Takeaway. The paper’s amplification relies on the *interaction* of:

heterogeneity in match quality (z) $\times$ on-the-job search ($\lambda_e > 0$).

8 Inspection-goods economy ($\alpha = 1$): why amplification disappears

When $\alpha = 1$, the signal perfectly reveals match quality at the meeting stage. Then search becomes closer to “shopping for the right match”.

8.1 Figures 5–7: impulse responses are small

Fig. 5 (flows and unemployment). The model predicts:

- a modest increase in UE,
- little movement in EE (because the shock does not change relative ranking of match qualities),
- a small decline in unemployment.

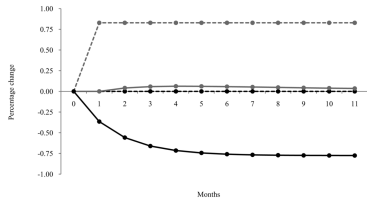


FIG. 5.—Inspection model, percentage change of the UE rate (dashed black line), the EU rate (dashed grey line), the EE rate (solid grey line), and the unemployment rate (solid black line) in response to a 1 percent increase in y .

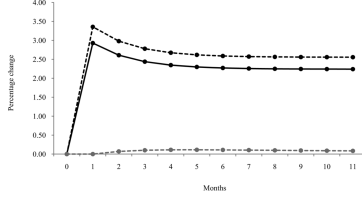


FIG. 6.—Inspection model, percentage change of total vacancies (solid black line), vacancies for unemployed workers (dashed black line), and vacancies for employed workers (dashed grey line) in response to a 1 percent increase in y .

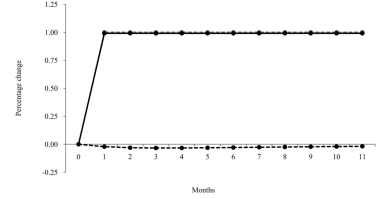


FIG. 7.—Inspection model, percentage change of the aggregate component of productivity (dashed grey line), the average idiosyncratic component of productivity (dashed black line), and the average labor productivity (solid black line) in response to a 1 percent increase in y .

Fig. 6 (vacancies). Vacancy creation responds mostly in the unemployed market; vacancies targeted at employed workers barely move.

Fig. 7 (productivity decomposition). Average match quality $\bar{z}$ changes little; average labor productivity rises almost one-for-one with y .

8.2 Analytic intuition: why the UE response is limited

A key elasticity decomposition (paper Eqs. (19)–(21)) implies that, once γ is pinned down by an empirical elasticity of job finding with respect to v/u , the remaining amplification depends on how much v/u itself responds to y .

In the inspection model, v/u is relatively insensitive to y because the economy does not generate a strong procyclical “poaching” vacancy component. Hence the UE rate and unemployment respond only weakly.

9 Short summary

1. **Efficient directed search can still generate large labor-market fluctuations.** The model does not rely on inefficiency; it relies on equilibrium reallocation with on-the-job search.
2. **Experience vs inspection is crucial.** When match quality is learned only through experience ($\alpha = 0$), reallocation and vacancy composition strongly amplify shocks. When match quality is known immediately ($\alpha = 1$), amplification largely disappears.

3. **Vacancy composition is a core mechanism.** Cycles are not only about total vacancies, but also about *where* vacancies are posted (unemployed vs employed submarkets).

10 Conclusion

10.1 Summary

Menzio and Shi (2011) provide a directed-search model with **on-the-job search (OJS)** and **match-quality heterogeneity** that can generate **large business-cycle fluctuations in unemployment, vacancies, and worker flows** from **modest aggregate productivity shocks**, while preserving the property that search is **efficient**. The paper’s central message is that large labor-market volatility does *not* require search inefficiency or ad hoc wage rigidity; it can arise in an environment where workers and firms optimally reallocate across submarkets and across matches over the cycle.

The theoretical contribution is twofold:

1. The paper proves that the equilibrium is **unique and constrained efficient** (it decentralizes the planner allocation) in a directed-search environment.
2. The paper establishes **block recursivity**: equilibrium value and policy functions depend on the aggregate state *only through the aggregate productivity shock* (not the full distribution of workers across match qualities), which makes it feasible to compute non-steady-state dynamics with heterogeneity and OJS.

The empirical contribution is to **calibrate the model to match worker transition patterns** (UE, EU, and EE flows, plus tenure moments) and then simulate the model to measure the effect of aggregate productivity shocks. The key quantitative conclusion is stark:

- When matches are **experience goods** (match quality is learned mainly after working), productivity shocks generate the **observed comovement** between unemployment, vacancies, and transition rates and explain a **large fraction** of their observed volatility.
- When matches are **inspection goods** (match quality is effectively known at hiring), productivity shocks generate only a **negligible fraction** of labor-market volatility.

10.2 What the conclusion implies (interpretation)

The paper’s conclusion can be summarized as:

1. **OJS and heterogeneity are jointly essential.** Removing either element destroys the amplification mechanism: models without OJS and/or without match heterogeneity fail to replicate either the magnitude of unemployment/vacancy volatility or the empirical comovement of UE/EU/EE flows.
2. **The informational nature of matches matters for macro fluctuations.** Whether match quality is discovered only after employment (experience goods) versus observed at hiring (inspection goods) fundamentally changes how strongly the labor market responds to aggregate shocks.

3. **Efficiency is compatible with volatility.** Because the equilibrium is efficient, large cyclical fluctuations in unemployment and vacancies are not automatically evidence of inefficient matching; they may reflect efficient, shock-driven *reallocation* across matches and submarkets.

10.3 Economic intuition: why the model amplifies small productivity shocks

The amplification mechanism is best understood as a **reallocation-and-composition channel** operating through OJS and vacancy composition.

Step 1: a small rise in aggregate productivity raises the surplus from upgrading matches. Match output is $y + z$ (aggregate y plus match quality z). When y increases, the gain from moving from a mediocre match to a better match becomes larger in present value terms. Thus, more employed workers in medium-quality matches find it worthwhile to search and upgrade.

Step 2: firms shift vacancy posting toward poaching employed workers. Because employed workers are now more attractive (they can be upgraded and are already productive), firms post more vacancies in submarkets targeting employed searchers. Hence the cycle is not just “more vacancies,” but “more *poaching* vacancies.”

Step 3: vacancy composition changes market tightness and worker flows. The rise in poaching vacancies raises job-to-job transitions (EE) and affects the entire flow system:

- EE increases strongly (procyclical upgrading),
- UE rises (job finding improves),
- EU falls (fewer separations / less destruction),
- unemployment declines substantially.

Even though EE does not mechanically move unemployment one-for-one, it changes the allocation of vacancies and the distribution of match values, which in turn feeds back into UE/EU and overall labor-market tightness.

Step 4: the model simultaneously keeps measured labor productivity relatively smooth. Average labor productivity can be decomposed as

$$\text{Average productivity} = y + \bar{z}.$$

In the experience-goods economy, a boom can reduce the endogenous destruction threshold (keeping more low- z matches alive), which lowers $\bar{z}$. That countercyclical movement in $\bar{z}$ partially offsets movements in y , so aggregate labor productivity remains comparatively smooth even when unemployment and vacancies move a lot. This joint pattern is exactly what motivates the paper: *small productivity volatility coexisting with large labor-market volatility*.

Why inspection goods kill amplification. If match quality is revealed at hiring, workers and firms can sort efficiently at entry. Then the main cyclical margin is a standard vacancy/unemployment tightness response, and the model does not generate a large procyclical “poaching” component. Without a strong vacancy-composition response and without large OJS-driven upgrading, UE and unemployment respond only weakly to y shocks.

10.4 Takeaway

Menzio and Shi (2011) show that an efficient directed-search labor market with on-the-job search and match heterogeneity can generate large business-cycle fluctuations through shock-induced reallocation and vacancy composition—but only when match quality is learned through experience rather than observed at hiring.